

INVESTMENTS

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CHAPTER 9

The Capital Asset Pricing Model

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

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Study Guide · Prepared for Personal Use

LO1 · The Capital Asset Pricing Model

The **CAPM** (published 1964–1966 by Sharpe, Lintner, and Mossin, building on Markowitz's 1952 portfolio theory) predicts the equilibrium relationship between an asset's risk and its expected return. It serves two vital functions: providing a benchmark "fair" return for evaluating investments, and helping estimate expected returns on assets not yet traded (e.g., an IPO or a new project).

The CAPM's Two Sets of Assumptions

- **Investor behaviour** — all investors are rational mean-variance optimizers with a common time horizon, using an identical input list (homogeneous expectations, consistent with all relevant information being public).
- **Market structure** — all assets are publicly traded; investors can borrow or lend at a common risk-free rate and take short positions; there are no taxes and no transaction costs.

These assumptions are admittedly strong — the chapter later devotes considerable attention (LO2) to how relaxing each one changes the model's predictions.

Why All Investors Hold the Market Portfolio

If every investor uses the same input list, every investor's Markowitz optimization produces the **same** efficient frontier and the same tangency (optimal risky) portfolio. Since all investors hold risky assets in identical proportions, and since lending and borrowing cancel out in aggregate (every lender is matched by a borrower), the aggregate risky holdings of all investors must equal the entire "market portfolio" — the value-weighted portfolio of every traded asset. The common tangency portfolio is therefore the market portfolio, M, and the CAL each investor uses becomes the **Capital Market Line (CML)** from Chapter 6.

If any traded asset were excluded from investors' optimal portfolios, its demand would fall to zero, its price would fall, and it would become progressively more attractive until it was priced to justify inclusion — in equilibrium, every asset must be held by someone, in exactly its value-weighted proportion.

The Passive Strategy Is Efficient: because the market portfolio is mean-variance efficient under CAPM assumptions, a passive investor holding a broad market-index fund (and mixing it with T-bills according to personal risk aversion) achieves an efficient outcome without any need for security analysis.

The Risk Premium of the Market Portfolio

In equilibrium, the reward-to-risk ratio must be equal across all assets and the market portfolio itself — otherwise investors would shift toward the better trade-off, moving prices until ratios equalize. This condition, applied to any individual security, is what pins down the CAPM's central relationship (see Formulas page).

Expected Return–Beta Relationship

Rearranging the equalized reward-to-risk condition yields the famous result: a security's risk premium equals its **beta** (its proportional contribution to market-portfolio variance) times the market portfolio's own risk premium. Total expected return is therefore the risk-free rate (compensation for waiting) plus a risk premium (compensation for worrying) that depends **only** on systematic risk — *not* on an asset's total volatility. A speculative biotech stock might have enormous total variance, but if most of that variance is firm-specific (diversifiable), its CAPM-required risk premium can still be modest.

The Security Market Line (SML)

The expected return–beta relationship, plotted graphically, is the **security market line (SML)**. Its slope is the market risk premium, and (unlike the CML, which only prices well-diversified portfolios) the SML applies to **every** individual asset, priced or mispriced, diversified or not — because beta, not total risk, is what the SML plots against expected return.

The vertical distance between a security's actual expected return and its "fair" SML-predicted return is its **alpha (α)**. Underpriced securities plot *above* the SML (positive alpha); overpriced securities plot *below* it (negative alpha). Security analysis, in this framework, is fundamentally about hunting for non-zero alphas — exactly the input the Treynor-Black active-portfolio model (Chapter 8) requires.

The CAPM also has practical uses beyond portfolio choice: setting a project's **hurdle rate** in capital budgeting, and determining a "fair" allowed rate of return for regulated utilities (see the worked example on the Examples page).

LO2 · Assumptions and Extensions of the CAPM

Relaxing each restrictive CAPM assumption produces a richer, more realistic (but more complex) model:

Short-Sale Impediments

Short selling is harder than going long: potential losses (and thus required collateral) are unlimited, shares available to borrow are limited, and many institutions (and Canadian/U.S. regulation) restrict short sales. Because impeded short-selling weakens the natural correction of overpriced securities, it can allow price bubbles to persist longer than the frictionless CAPM would predict.

Taxes

Different investors face different effective tax rates on the same security's returns, which could in principle produce different optimal portfolios for different tax clienteles. In practice, no decisive evidence shows taxes to be a major driver of returns — plausibly because tax clienteles self-select (high-bracket investors avoid high-dividend stocks; tax-exempt investors take up the resulting bargains), neutralizing the effect in aggregate.

Restricted Borrowing — The Zero-Beta Model

If investors cannot borrow freely at the risk-free rate, lenders and borrowers reach different tangency portfolios, and the market portfolio is no longer necessarily each investor's optimal risky portfolio. Fischer Black showed that every efficient portfolio (except the global minimum-variance portfolio) has a "zero-beta companion" portfolio Z uncorrelated with it. Substituting the return on Z for the risk-free rate produces a CAPM-like equation (Formulas page, F2) with a **flatter SML** than the simple model — since risk-tolerant investors who cannot leverage instead tilt toward high-beta stocks, bidding up their prices and compressing their risk premiums.

Labour Income and Non-Traded Assets

Not all wealth is tradeable — human capital (the present value of future labour income) and private businesses are both large, untraded components of total wealth, often exceeding the value of all publicly traded assets combined. Investors whose labour income (or private business) is highly correlated with a particular sector should rationally avoid further exposure to that sector (e.g., an airline employee avoiding airline stocks), which can cause optimal portfolios to deviate from the simple mean-beta prediction.

A Multifactor CAPM and Liquidity

Investors also demand compensation for exposure to non-market risks that affect their ability to consume as planned (e.g., changing future investment opportunities — Merton's Intertemporal CAPM) and for **liquidity risk**: securities whose returns fall when overall market liquidity dries up should offer a liquidity premium. Empirically, the liquidity premium appears to be of a similar order of magnitude to the ordinary market risk premium — large enough to be a first-order consideration in asset pricing, not a minor footnote.

LO3 · The CAPM and the Academic World

Testing the CAPM is genuinely difficult, for a fundamental reason: the theory's market portfolio should include **all** risky assets in the economy — not just traded stocks, but real estate, human capital, and private businesses — none of which can be fully observed or measured. This is known as Roll's critique: since the true market portfolio is unobservable, the CAPM's central prediction (that the market portfolio is mean-variance efficient) is not directly testable using any proxy index alone.

Multi-index and intertemporal extensions of the CAPM attempt to capture additional priced risk factors (interest rates, inflation, business-cycle shifts). But because it is often unclear *which* fundamental uncertainties a candidate factor is really hedging, many researchers remain skeptical of empirically driven (rather than theory-driven) factor selection. In the academic world, the single-index CAPM is now generally viewed as too simple, while no single successor model has yet achieved consensus acceptance.

LO4 · The CAPM and the Investment Industry

While academics search for richer multifactor models, industry practice has stayed largely anchored to the single-index CAPM. One reason: although the CAPM's claim that the market portfolio is efficient cannot be directly tested, it has become increasingly clear that **consistently beating even a fairly narrow benchmark index** (such as the S&P 500) has proven extremely difficult for professional money managers.

Burton Malkiel's influential study of equity mutual fund alphas (1971–1991) found a roughly bell-shaped distribution of fund alphas centred slightly negative but statistically indistinguishable from zero — professional managers, on average, do *not* outperform the market index on a risk-adjusted basis. This finding has been reconfirmed repeatedly by subsequent research.

Given how hard the passive benchmark is to beat, an index portfolio is treated as "efficient for all practical purposes" across several industry applications: (1) a diversification vehicle to combine with an actively managed component (the Treynor-Black approach), (2) a benchmark for evaluating and compensating portfolio managers, (3) a reference point in legal disputes over "fair" investment compensation, and (4) a tool for setting fair prices in regulated industries.

Chapter 9 — Big-Picture Takeaways

- Under CAPM assumptions, every investor holds the same optimal risky portfolio — the market portfolio — combined with the risk-free asset according to individual risk aversion.
- A security's risk premium depends only on its beta (systematic risk), not on its total volatility — firm-specific risk earns no compensation because it can be diversified away.

- The SML plots the expected return–beta relationship for any asset; positive-alpha securities plot above it, negative-alpha securities below.
- Relaxing CAPM's restrictive assumptions (short-sale limits, taxes, restricted borrowing, non-traded assets, liquidity) enriches the model but complicates its clean single-beta prediction.
- Because the true "market portfolio" of all assets (including human capital and private businesses) is unobservable, the core CAPM prediction is not directly testable — Roll's critique.
- Despite this, professional managers have historically struggled to beat passive index benchmarks on a risk-adjusted basis, which is why the industry continues to rely heavily on CAPM-based passive benchmarking.

The expected return–beta relationship (F1) is the single most important formula in the course to this point.

F1 | Expected Return–Beta Relationship (CAPM / SML)

$$E(R_i) = \beta_i \times E(R_M)$$

$$\text{i.e. } E(r_i) - r_f = \beta_i \times [E(r_M) - r_f]$$

$$\text{where } \beta_i = \text{Cov}(r_i, r_M) / \sigma_M^2$$

R denotes excess return over r_f . Beta measures a security's contribution to market-portfolio variance, as a fraction of total market variance. This is the formal statement of the SML.

F2 | Zero-Beta CAPM (Restricted Borrowing)

$$E(r_i) = E(r_Z) + \beta_i \times [E(r_M) - E(r_Z)]$$

$E(r_Z)$ = expected return on the zero-beta companion portfolio of M, replacing r_f . Produces a flatter SML than the simple CAPM, since $E(r_Z) > r_f$.

F3 | Alpha (Mispricing Relative to the SML)

$$\alpha_i = E(r_i)_{\text{actual}} - [r_f + \beta_i \times (E(r_M) - r_f)]$$

*Positive alpha → security plots above the SML (underpriced, offers better-than-fair return for its risk).
Negative alpha → plots below (overpriced).*

F4 | CAPM Hurdle Rate / Regulated Utility Rate-of-Return

$$\text{Required Return} = r_f + \beta \times (\text{Market Risk Premium})$$

$$\text{Fair Profit} = \text{Required Return} \times \text{Invested Capital}$$

Applies the SML to set a project's minimum acceptable return, or a regulated utility's allowed profit on its invested capital base.

EXAMPLE | Computing a Stock's Alpha from the SML

Given: The market's expected return is 14%, the T-bill rate is 6%, and a stock has beta = 1.2. An analyst independently forecasts the stock's expected return at 17%.

Step 1 — SML-predicted ("fair") return:

$$r_{\text{fair}} = 6 + 1.2 \times (14 - 6) = 6 + 9.6 = 15.6\%$$

Step 2 — Alpha:

$$\alpha = 17\% - 15.6\% = 1.4\%$$

Because the analyst's forecast (17%) exceeds the SML-fair return (15.6%), the stock plots above the SML — it appears underpriced, with a positive alpha of 1.4%. This is exactly the kind of security that would earn a positive weight in a Treynor-Black active portfolio (Chapter 8).

EXAMPLE 9.1 | Using the CAPM — Utility Rate-Making

Given: Equityholders have invested \$100 million in a regulated utility. The equity beta is 0.6. The T-bill rate is 6% and the market risk premium is 8%.

Step 1 — Fair required return:

$$\text{Required return} = 6 + 0.6 \times 8 = 6 + 4.8 = 10.8\%$$

Step 2 — Fair total profit:

$$\text{Fair profit} = 10.8\% \times \$100 \text{ million} = \$10.8 \text{ million}$$

The regulator would allow the utility to set prices expected to generate \$10.8 million in annual profit — enough to fairly compensate equityholders for the systematic risk they bear ($\beta = 0.6$), but no more.

PRACTICE PROBLEM | Glacier Point Advisors

Scenario

You are an analyst at Glacier Point Advisors in Calgary. The S&P/TSX Composite is expected to return 11% next year; the T-bill rate is 3%. You are evaluating three stocks for a client's watch list:

Stock	Beta	Analyst's Forecast $E(r)$	
RockCrest Mining	1.4	16.0%	
Foothills Utilities	0.5	7.5%	
Chinook Retail	1.0	12.5%	

Part A — The SML and Alpha

- Calculate the SML-predicted ("fair") expected return for each of the three stocks.
- Calculate each stock's alpha.
- Which stock(s) appear underpriced, and which appear overpriced? Explain using the SML.

Part B — Portfolio Construction Implications

- Based on Part A, which stock would you expect to receive a positive weight in an active portfolio (Chapter 8 framework), and which a negative (short) weight?
- Chinook Retail has a beta of exactly 1.0 and an alpha of approximately zero. What does this imply about how Chinook behaves relative to the passive market-index portfolio?

Part C — CAPM Application to Capital Budgeting

Foothills Utilities is a regulated Alberta utility. Equityholders have invested \$250 million in the firm, and its equity beta is 0.5 (as above).

- Using the CAPM, calculate the fair required rate of return for Foothills' equityholders.
- Calculate the fair total annual profit the utility should be allowed to earn.
- If the regulator instead used Foothills' total return volatility (not beta) to set the allowed return, explain briefly why this would be inconsistent with CAPM logic.

Part D — Relaxing CAPM Assumptions

Suppose short-selling RockCrest Mining shares is very difficult in practice (few shares available to borrow).

- (i) Explain, referencing LO2, why this impediment could allow RockCrest's price to become and remain overvalued for longer than the frictionless CAPM would predict.
- (ii) A Glacier Point client's personal income is heavily tied to the Alberta energy sector. Explain why this client might rationally hold a different optimal portfolio than the CAPM's prediction, referencing the "non-traded assets" extension.

Hints & Guidance

Part A tests LO1 and Formulas F1 & F3 (mirror the Alpha example exactly). Part B synthesizes LO1 with the Chapter 8 active-portfolio framework. Part C tests LO1 and Formula F4 (mirror Example 9.1 exactly). Part D tests LO2 — look back to the short-sale-impediments and non-traded-assets subsections.